

Assignment 2

Install and configure pfSense Firewall, Squid, and SquidGuard.

1) Boot pfSense firewall successfully.

```
Starting syslog...done.
Starting CRON... done.
pfSense 2.8.1-RELEASE amd64 20260513-1943
Bootup complete

FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)

VMware Virtual Machine - Netgate Device ID: 0015e5e7939f993b13ba

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WAN (wan) -> em0 -> v4/DHCP4: 192.168.74.215/24
LAN (lan) -> em1 -> v4: 192.168.1.1/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults         13) Update from console
5) Reboot system                    14) Enable Secure Shell (sshd)
6) Halt system                      15) Restore recent configuration
7) Ping host                        16) Restart PHP-FPM
8) Shell

Enter an option: █
```

2) Configure WAN interface to obtain IPv4 address using DHCP.

```
1 - WAN (em0 - dhcp, dhcp6)
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure: 1

Configure IPv4 address WAN interface via DHCP? (y/n) y

Configure IPv6 address WAN interface via DHCP6? (y/n) n

Enter the new WAN IPv6 address. Press <ENTER> for none:
>
Disabling IPv4 DHCPD...
Disabling IPv6 DHCPD...

Do you want to revert to HTTP as the webConfigurator protocol? (y/n) n

Please wait while the changes are saved to WAN...
Reloading filter...
Reloading routing configuration...
DHCPD...

The IPv4 WAN address has been set to dhcp

Press <ENTER> to continue. █
```

3) Select LAN interface for IP configuration.

```
VMware Virtual Machine - Netgate Device ID: 0015e5e7939f993b13ba

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

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8) Shell

Enter an option: 2

Available interfaces:

1 - WAN (em0 - dhcp)
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure: █
```

4) Assign static LAN IP address 192.168.25.5/24.

```
Enter an option: 2

Available interfaces:

1 - WAN (em0 - dhcp)
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure: 2

Configure IPv4 address LAN interface via DHCP? (y/n) n

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.25.5

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
> █
```

5) Disable DHCP server and IPv6 on LAN interface.

```
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via DHCP6? (y/n) n

Enter the new LAN IPv6 address. Press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? (y/n) n
Disabling IPv4 DHCPD...
Disabling IPv6 DHCPD...

Do you want to revert to HTTP as the webConfigurator protocol? (y/n) n

Please wait while the changes are saved to LAN...
Reloading filter...
Reloading routing configuration...
DHCPD...

The IPv4 LAN address has been set to 192.168.25.5/24
You can now access the webConfigurator by opening the following URL in your web
browser:
        https://192.168.25.5/

Press <ENTER> to continue. █
```

6) Verify LAN IP configuration and access URL (https://192.168.25.5).

```
The IPv4 LAN address has been set to 192.168.25.5/24
You can now access the webConfigurator by opening the following URL in your web
browser:
        https://192.168.25.5/

Press <ENTER> to continue.
VMware Virtual Machine - Netgate Device ID: 0015e5e7939f993b13ba

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WAN (wan) -> em0 -> v4/DHCP4: 192.168.74.215/24
LAN (lan) -> em1 -> v4: 192.168.25.5/24

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5) Reboot system                   14) Enable Secure Shell (sshd)
6) Halt system                     15) Restore recent configuration
7) Ping host                       16) Restart PHP-FPM
8) Shell

Enter an option: █
```

7) Reset pfSense admin account and set new password.

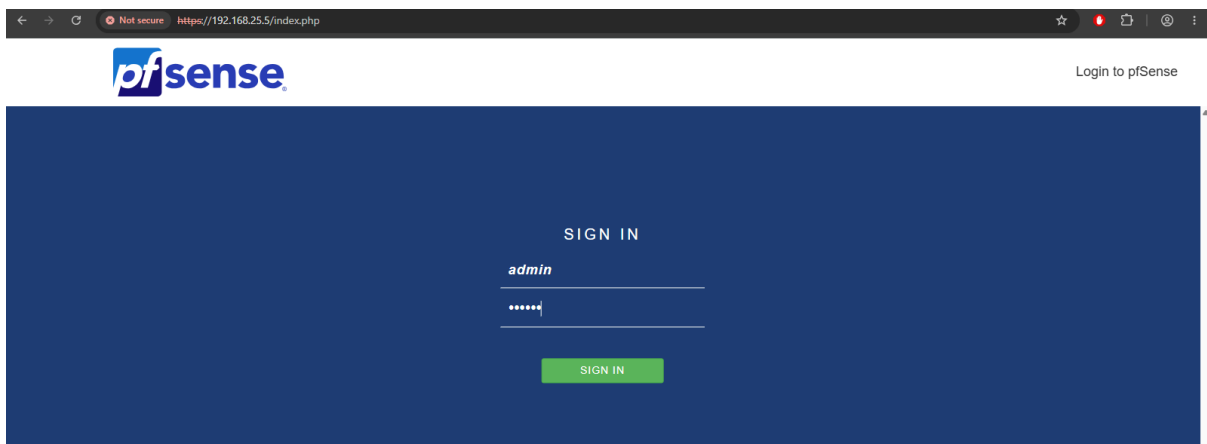
```
0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                   14) Enable Secure Shell (sshd)
6) Halt system                     15) Restore recent configuration
7) Ping host                       16) Restart PHP-FPM
8) Shell

Enter an option: 3

The authentication configuration and privileges for the "admin" account will be
reset to the default.
Proceed? (y/n): y

The default administrator account in the User Manager ("admin") has been reset.
The password must now be set to a new value.
Changing password for 'admin' in the User Manager database.
Hints:
* Current NIST guidelines prioritize password length over complexity.
* The password cannot be identical to the username.
New Password:
Confirm New Password: █
```

8) Login to pfSense WebConfigurator using admin credentials.



Not secure https://192.168.25.5/index.php

pfSense Login to pfSense

SIGN IN

admin

.....

SIGN IN

9) Run initial setup wizard and configure hostname/domain.

The screenshot shows the 'General Information' step of the pfSense Setup Wizard. The breadcrumb trail is 'Wizard / pfSense Setup / General Information'. A progress bar indicates 'Step 2 of 9'. The page title is 'General Information'. Below the title, a message states: 'On this screen the general pfSense parameters will be set.' The form contains the following fields and options:

- Hostname:** A text input field containing 'pfSense'. Below it, a note says 'Name of the firewall host, without domain part.' and examples: 'pfsense, firewall, edgefw'.
- Domain:** A text input field containing 'home.arpa'. Below it, a note says 'Domain name for the firewall.' and examples: 'home.arpa, example.com'. A detailed note explains: 'Do not end the domain name with '.local' as the final part (Top Level Domain, TLD). The '.local' TLD is widely used by mDNS (e.g. Avahi, Bonjour, Rendezvous, Airprint, Airplay) and some Windows systems and networked devices. These will not network correctly if the router uses '.local' as its TLD. Alternatives such as 'home.arpa', 'local.lan', or 'mylocal' are safe.'
- Primary DNS Server:** An empty text input field.
- Secondary DNS Server:** An empty text input field.
- Override DNS:** A checked checkbox. Below it, a note says 'Allow DNS servers to be overridden by DHCP/PPP on WAN'.

At the bottom of the form is a blue button labeled '>> Next'.

10) Configure NTP time server and select timezone.

The screenshot shows the 'Time Server Information' step of the pfSense Setup Wizard. The breadcrumb trail is 'Wizard / pfSense Setup / Time Server Information'. A progress bar indicates 'Step 3 of 9'. The page title is 'Time Server Information'. Below the title, a message states: 'Please enter the time, date and time zone.' The form contains the following fields and options:

- Time server hostname:** A text input field containing '2.pfsense.pool.ntp.org'. Below it, a note says 'Enter the hostname (FQDN) of the time server.'
- Timezone:** A dropdown menu with 'Asia/Kolkata' selected.

At the bottom of the form is a blue button labeled '>> Next'.

11) Confirm LAN IP address and subnet mask settings.

The screenshot shows the 'Configure LAN Interface' step of the pfSense Setup Wizard. The breadcrumb trail is 'Wizard / pfSense Setup / Configure LAN Interface'. A progress bar indicates 'Step 5 of 9'. The main heading is 'Configure LAN Interface'. Below it, a message states: 'On this screen the Local Area Network information will be configured.' There are two input fields: 'LAN IP Address' with the value '192.168.25.5' and a note 'Type dhcp if this interface uses DHCP to obtain its IP address.', and 'Subnet Mask' with the value '24'. A blue 'Next' button is at the bottom.

12) Reload pfSense configuration and apply changes.

The screenshot shows the 'Reload configuration' step of the pfSense Setup Wizard. The breadcrumb trail is 'Wizard / pfSense Setup / Reload configuration'. A progress bar indicates 'Step 7 of 9'. The main heading is 'Reload configuration'. Below it, a message states: 'Click 'Reload' to reload pfSense with new changes.' A blue 'Reload' button is at the bottom.

13) Complete setup wizard and verify successful installation.

The screenshot shows the 'Wizard completed' step of the pfSense Setup Wizard. The breadcrumb trail is 'Wizard / pfSense Setup / Wizard completed.'. A progress bar indicates 'Step 9 of 9'. The main heading is 'Wizard completed.'. Below it, a message states: 'Congratulations! pfSense is now configured.' followed by a recommendation to check for updates and a 'Check for updates' button. Then, a message states: 'Remember, we're here to help.' followed by a link to support services. Next, a 'User survey' section asks for feedback with a link to the 'Anonymous User Survey'. Finally, a 'Useful resources' section lists links to the website, store, forum, and newsletter. A blue 'Finish' button is at the bottom.

14) Verify pfSense dashboard status and network interface configuration.

The screenshot displays the pfSense dashboard. The top navigation bar includes links for System, Interfaces, Firewall, Services, VPN, Status, Diagnostics, and Help. The main content area is divided into two panels. The left panel, titled 'System Information', shows details about the user (admin@192.168.25.1), system (VMware Virtual Machine), BIOS (Phoenix Technologies LTD), version (2.8.1-RELEASE), CPU type (Intel(R) Core(TM) i7-7700 CPU @ 3.60GHz), hardware crypto (Inactive), kernel PTI (Enabled), MDS Mitigation (Inactive), uptime (00 Hour 27 Minutes 00 Second), current date/time (Thu Jun 4 17:13:51 IST 2026), DNS server(s) (127.0.0.1, ::1, 192.168.72.20), and last config change (Thu Jun 4 17:10:25 IST 2026). The right panel, titled 'Netgate Services And Support', shows the contract type (Community Support) and provides links to various support resources, including the Netgate Resource Library, Upgrade Your Support, Netgate Global Support FAQ, Netgate Professional Services, Community Support Resources, Official pfSense Training by Netgate, and Visit Netgate.com. A note mentions that purchasing a Netgate Global TAC Support subscription requires a Netgate Device ID (NDI).

System Information

User: admin@192.168.25.1 (Local Database)

System: VMware Virtual Machine
Netgate Device ID: 0015e5e7939f993b13ba

BIOS: Vendor: Phoenix Technologies LTD
Version: 6.00
Release Date: Mon Mar 24 2025
Boot Method: BIOS

Version: 2.8.1-RELEASE (amd64)
built on Thu May 14 1:13:00 IST 2026
FreeBSD 15.0-CURRENT

The system is on the latest version.
Version information updated at: Thu Jun 4 16:48:33 IST 2026

CPU Type: Intel(R) Core(TM) i7-7700 CPU @ 3.60GHz
2 CPUs : 2 package(s) x 1 core(s)
AES-NI CPU Crypto: Yes (inactive)
QAT Crypto: No

Hardware crypto: Inactive

Kernel PTI: Enabled

MDS Mitigation: Inactive

Uptime: 00 Hour 27 Minutes 00 Second

Current date/time: Thu Jun 4 17:13:51 IST 2026

DNS server(s):

- 127.0.0.1
- ::1
- 192.168.72.20

Last config change: Thu Jun 4 17:10:25 IST 2026

Netgate Services And Support

Contract type: Community Support
Community Support Only

NETGATE AND pfSense COMMUNITY SUPPORT RESOURCES

If you purchased your pfSense gateway firewall appliance from Netgate and elected Community Support at the point of sale or installed pfSense on your own hardware, you have access to various community support resources. This includes the [NETGATE RESOURCE LIBRARY](#).

You also may upgrade to a Netgate Global Technical Assistance Center (TAC) Support subscription. We're always on! Our team is staffed 24x7x365 and committed to delivering enterprise-class, worldwide support at a price point that is more than competitive when compared to others in our space.

- Upgrade Your Support
- Netgate Global Support FAQ
- Netgate Professional Services
- Community Support Resources
- Official pfSense Training by Netgate
- Visit Netgate.com

If you decide to purchase a Netgate Global TAC Support subscription, you **MUST** have your Netgate Device ID (NDI) from your firewall in order to validate support for this unit. Write down your NDI and store it in a safe place. You can purchase TAC supports [here](#).

Interfaces

Interface	Speed	Duplex	IP Address
WAN	1000baseT	<full-duplex>	192.168.74.215
LAN	1000baseT	<full-duplex>	192.168.25.5

15) Search and install Squid, SquidGuard, and Lightsquid packages.

The screenshot displays the pfSense Package Manager interface. The top navigation bar includes links for System, Interfaces, Firewall, Services, VPN, Status, Diagnostics, and Help. The main content area is titled 'System / Package Manager / Available Packages'. Below the title, there are tabs for 'Installed Packages' and 'Available Packages'. A search bar is present with the search term 'squid'. Below the search bar, there is a table of available packages. The table has columns for Name, Version, and Description. Three packages are listed: Lightsquid (version 3.0.7.5), squid (version 0.5.3), and squidGuard (version 1.16.23). Each package has an 'Install' button. The description for Lightsquid mentions it is a high performance web proxy reporting tool. The description for squid mentions it is a high performance web proxy cache. The description for squidGuard mentions it is a high performance web proxy URL filter. Package dependencies are listed for each package.

System / Package Manager / Available Packages

Installed Packages Available Packages

Search

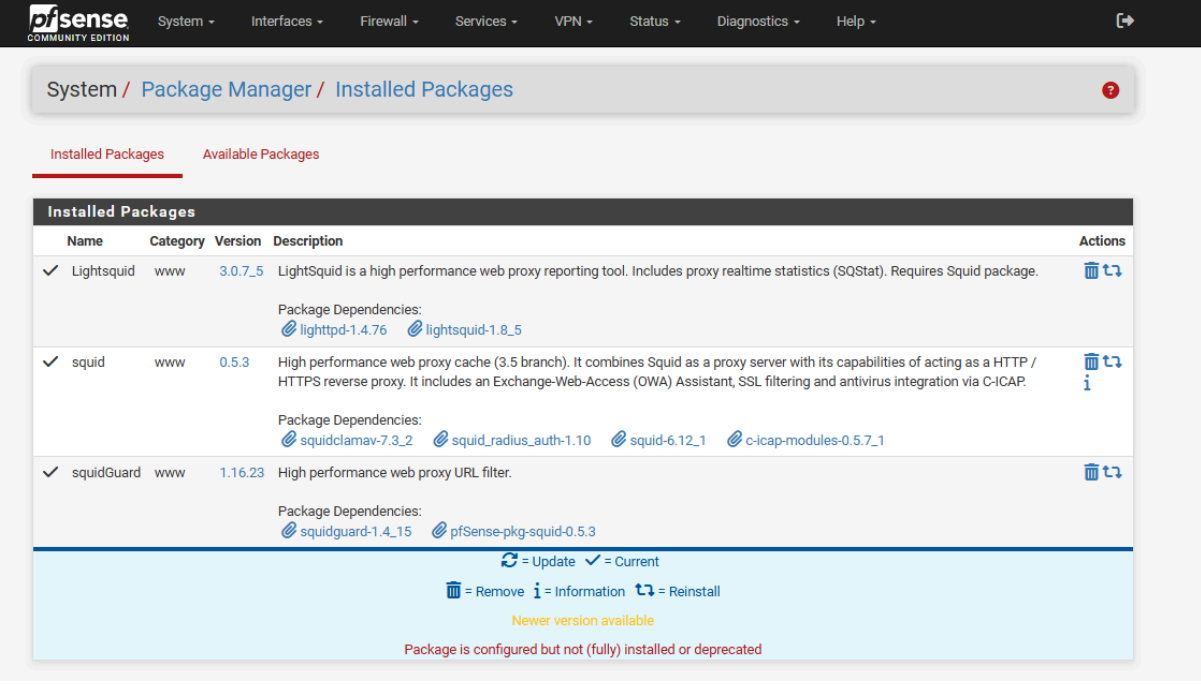
Search term: squid Both Search Clear

Enter a search string or *nix regular expression to search package names and descriptions.

Packages

Name	Version	Description	Install
Lightsquid	3.0.7.5	LightSquid is a high performance web proxy reporting tool. Includes proxy realtime statistics (SQStat). Requires Squid package. Package Dependencies: lighttpd-1.4.76 lightsquid-1.8.5	+ Install
squid	0.5.3	High performance web proxy cache (3.5 branch). It combines Squid as a proxy server with its capabilities of acting as a HTTP / HTTPS reverse proxy. It includes an Exchange-Web-Access (OWA) Assistant, SSL filtering and antivirus integration via C-ICAP. Package Dependencies: squidclamav-7.3.2 squid_radius_auth-1.10 squid-6.12.1 c-icap-modules-0.5.7.1	+ Install
squidGuard	1.16.23	High performance web proxy URL filter. Package Dependencies: squidguard-1.4.15 pfSense-pkg-squid-0.5.3	+ Install

16) Verify successful installation of proxy-related packages.

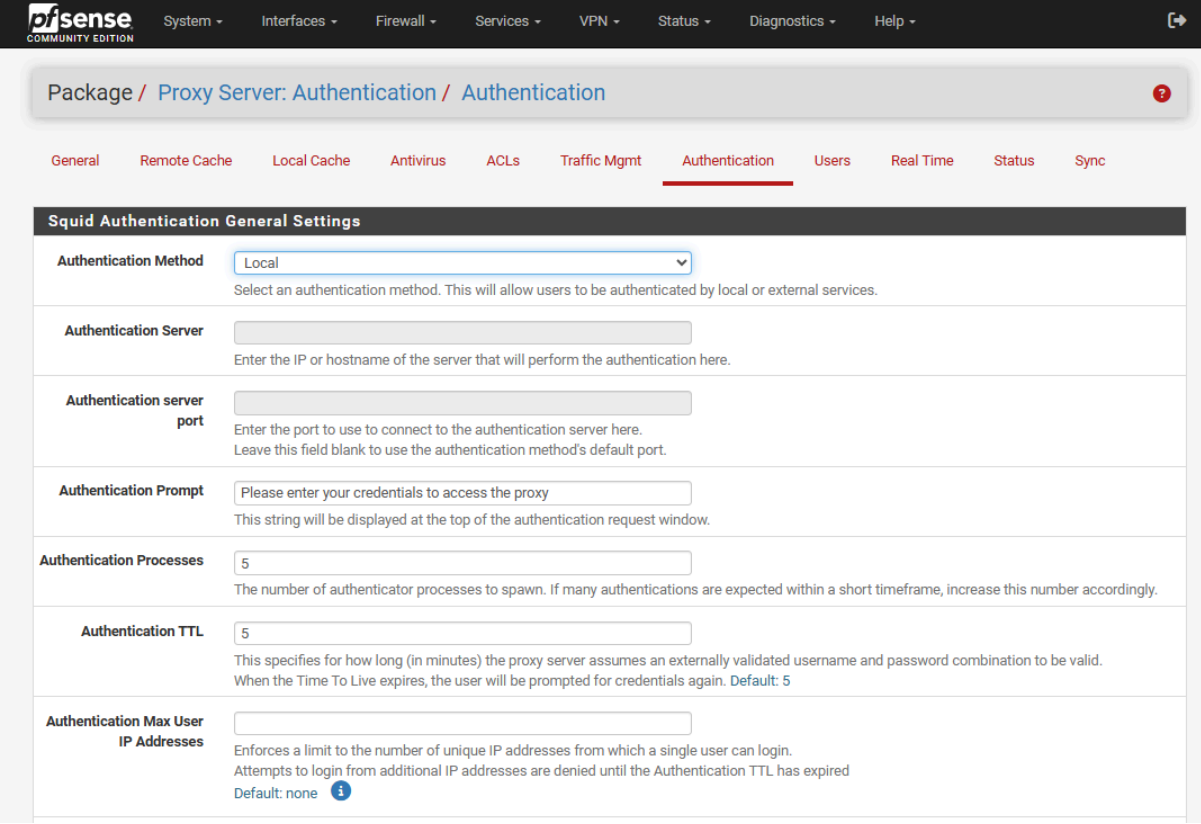


The screenshot shows the pfSense Package Manager interface. The breadcrumb trail is System / Package Manager / Installed Packages. There are two tabs: 'Installed Packages' (active) and 'Available Packages'. The 'Installed Packages' table lists three packages:

Name	Category	Version	Description	Actions
✓ Lightsquid	www	3.0.7.5	LightSquid is a high performance web proxy reporting tool. Includes proxy realtime statistics (SQStat). Requires Squid package. Package Dependencies: lighttpd-1.4.76 lightsquid-1.8.5	Remove, Update, Reinstall
✓ squid	www	0.5.3	High performance web proxy cache (3.5 branch). It combines Squid as a proxy server with its capabilities of acting as a HTTP / HTTPS reverse proxy. It includes an Exchange-Web-Access (OWA) Assistant, SSL filtering and antivirus integration via C-ICAP. Package Dependencies: squidclamav-7.3.2 squid_radius_auth-1.10 squid-6.12_1 c-icap-modules-0.5.7_1	Remove, Update, Reinstall, Info
✓ squidGuard	www	1.16.23	High performance web proxy URL filter. Package Dependencies: squidguard-1.4.15 pfSense-pkg-squid-0.5.3	Remove, Update, Reinstall

Legend:
↻ = Update ✓ = Current
🗑 = Remove ⓘ = Information ↺ = Reinstall
Newer version available
Package is configured but not (fully) installed or deprecated

17) Configure Squid local user authentication settings.



The screenshot shows the pfSense Proxy Server: Authentication / Authentication settings page. The breadcrumb trail is Package / Proxy Server: Authentication / Authentication. The 'Authentication' tab is selected. The 'Squid Authentication General Settings' section contains the following fields:

- Authentication Method:** Local (dropdown menu). Select an authentication method. This will allow users to be authenticated by local or external services.
- Authentication Server:** (text input). Enter the IP or hostname of the server that will perform the authentication here.
- Authentication server port:** (text input). Enter the port to use to connect to the authentication server here. Leave this field blank to use the authentication method's default port.
- Authentication Prompt:** Please enter your credentials to access the proxy (text input). This string will be displayed at the top of the authentication request window.
- Authentication Processes:** 5 (text input). The number of authenticator processes to spawn. If many authentications are expected within a short timeframe, increase this number accordingly.
- Authentication TTL:** 5 (text input). This specifies for how long (in minutes) the proxy server assumes an externally validated username and password combination to be valid. When the Time To Live expires, the user will be prompted for credentials again. Default: 5
- Authentication Max User IP Addresses:** (text input). Enforces a limit to the number of unique IP addresses from which a single user can login. Attempts to login from additional IP addresses are denied until the Authentication TTL has expired. Default: none ⓘ

18) Configure Squid cache management and local cache parameters.

The screenshot shows the 'Local Cache' tab under 'Proxy Server: Cache Management'. The 'Squid Cache General Settings' section includes the following options:

- Disable Caching:** ☐ Disable caching completely. This may be required if Squid is only used as a proxy to audit website access.
- Cache Replacement Policy:** Heap LFUDA. The cache replacement policy decides which objects will remain in cache and which objects are replaced to create space for the new objects. Default: heap LFUDA.
- Low-Water Mark in %:** 90. The low-water mark for AUFS/UPS/diskd cache object eviction by the cache_replacement_policy algorithm.
- High-Water Mark in %:** 95. The high-water mark for AUFS/UPS/diskd cache object eviction by the cache_replacement_policy algorithm.
- Do Not Cache:** (Empty text area)

19) Enable Squid proxy service and select listening interfaces.

The screenshot shows the 'General' tab under 'Proxy Server: General Settings'. The 'Squid General Settings' section includes the following options:

- Enable Squid Proxy:** ☐ Check to enable the Squid proxy. **Important:** If unchecked, ALL Squid services will be disabled and stopped.
- Keep Settings/Data:** ☒ If enabled, the settings, logs, cache, AV defs and other data will be preserved across package reinstalls. **Important:** If disabled, all settings and data will be wiped on package uninstall/reinstall/upgrade.
- Listen IP Version:** IPv4. Select the IP version Squid will use to select addresses for accepting client connections.
- CARP Status VIP:** none. Used to determine the HA MASTER/BACKUP status. Squid will be stopped when the chosen VIP is in BACKUP status, and started in MASTER status. **Important:** Don't forget to generate Local Cache on the secondary node and configure XMLRPC Sync for the settings synchronization.
- Proxy Interface(s):** WAN, LAN, loopback. The interface(s) the proxy server will bind to. Use CTRL + click to select multiple interfaces.
- Outgoing Network Interface:** Default (auto). The interface the proxy server will use for outgoing connections.
- Proxy Port:** 3128. This is the port the proxy server will listen on. Default: 3128.
- ICP Port:** (Empty text area)

20) Configure proxy server hostname, email, language, and header options.

The screenshot shows the 'Headers Handling, Language and Other Customizations' configuration page in pfSense. The page has a dark header with the title. Below the header, there are several configuration fields:

- Visible Hostname:** A text input field containing 'pfsensesrv'. Below it, a note says 'This is the hostname to be displayed in proxy server error messages.'
- Administrator's Email:** A text input field containing 'admin@localhost'. Below it, a note says 'This is the email address displayed in error messages to the users.'
- Error Language:** A dropdown menu set to 'en'. Below it, a note says 'Select the language in which the proxy server will display error messages to users.'
- X-Forwarded Header Mode:** A dropdown menu set to '(on)'. Below it, a note says 'Choose how to handle X-Forwarded-For headers. Default: on' with an information icon.
- Disable VIA Header:** A checkbox that is unchecked. Below it, a note says 'If not set, Squid will include a Via header in requests and replies as required by RFC2616.'
- URI Whitespace Characters Handling:** A dropdown menu set to 'strip'. Below it, a note says 'Choose how to handle whitespace characters in URL. Default: strip' with an information icon.
- Suppress Squid Version:** A checkbox that is unchecked. Below it, a note says 'Suppresses Squid version string info in HTTP headers and HTML error pages if enabled.'

At the bottom of the form, there are two buttons: 'Save' and 'Show Advanced Options'.

Perform the following tasks:

1. Create a user named ditiss

Allow access only to:

- CentOS website
- Red Hat website
- Amazon websites
- Microsoft Cloud websites
- Kali Linux website

The screenshot shows the 'Proxy Server: Local Users / Edit / Users' configuration page in pfSense. The page has a dark header with the pfSense logo and navigation tabs: System, Interfaces, Firewall, Services, VPN, Status, Diagnostics, and Help. Below the header, there is a breadcrumb trail: 'Proxy Server: Local Users / Edit / Users'. Below the breadcrumb, there are several tabs: General, Remote Cache, Local Cache, Antivirus, ACLs, Traffic Mgmt, Authentication, Users (which is selected), Real Time, Status, and Sync. Below the tabs, there is a section titled 'Squid Local Users'. This section contains three fields:

- Username:** A text input field containing 'ditiss'. Below it, a note says 'Enter the username here.'
- Password:** A password input field containing '*****'. Below it, a note says 'Enter the password here.'
- Description:** A text input field containing 'ditiss user'. Below it, a note says 'You may enter a description here for your reference (not parsed).'

At the bottom of the form, there is a 'Save' button.

Not secure https://192.168.25.5/pkg.php?xml=squid_users.xml

pfSense
COMMUNITY EDITION

System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

Package / Proxy Server: Local Users / Users

General Remote Cache Local Cache Antivirus ACLs Traffic Mgmt Authentication **Users** Real Time Status Sync

Username	Description	
ditiss	ditiss user	 
dbda	dbda user	 
		

pfSense
COMMUNITY EDITION

Proxy filter SquidGuard: Target categories / Edit / Target categories

General settings Common ACL Groups ACL **Target categories** Times Rewrites Blacklist Log

XMLRPC Sync

General Options

Name

Enter a unique name of this rule here.
The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter.

Order

Select the new position for this target category. Target categories are listed in this order on ACLs and are matched from the top down in sequence.

Domain List

centos.org www.centos.org redhat.com www.redhat.com amazon.com
www.amazon.com aws.amazon.com microsoft.com www.microsoft.com
azure.microsoft.com kali.org www.kali.org kali-linux.org
www.kali-linux.org

Enter destination domains or IP-addresses here. To separate them use space.
Example: mail.ru e-mail.ru yahoo.com 192.168.1.1

2. Create another user named dbda

Allow access only to:

- Python websites
- Oracle websites
- IBM websites
- Microsoft websites

Proxy filter SquidGuard: Target categories / Edit / Target categories 

[General settings](#) [Common ACL](#) [Groups ACL](#) [Target categories](#) [Times](#) [Rewrites](#) [Blacklist](#) [Log](#)

XMLRPC Sync

General Options

Name	common_whitelist Enter a unique name of this rule here. The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter.
Order	social_sites  Select the new position for this target category. Target categories are listed in this order on ACLs and are matched from the top down in sequence.
Domain List	google.com www.google.com gmail.com mail.google.com bing.com www.bing.com cdac.in www.cdac.in Enter destination domains or IP-addresses here. To separate them use space. Example: mail.ru e-mail.ru yahoo.com 192.168.1.1

Not securehttps://192.168.25.5/pkg.php?xml=squid_users.xml

pf

sense

COMMUNITY EDITION

System

Interfaces

Firewall

Services

VPN

Status

Diagnostics

Help

Package / Proxy Server: Local Users / Users

GeneralRemote CacheLocal CacheAntivirusACLsTraffic MgmtAuthenticationUsersReal TimeStatusSync

Username	Description	
ditiss	ditiss user	 
dbda	dbda user	 
		

Save

pf

sense

COMMUNITY EDITION

Proxy filter SquidGuard: Target categories / Edit / Target categories

General settingsCommon ACLGroups ACLTarget categoriesTimesRewritesBlacklistLog

XMLRPC Sync

General Options

Name

dbda_whitelist

Enter a unique name of this rule here.
The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter.

Order

ditiss_whitelist

Select the new position for this target category. Target categories are listed in this order on ACLs and are matched from the top down in sequence.

Domain List

python.orgwww.python.orgoracle.comwww.oracle.comibm.com
www.ibm.commicrosoft.comwww.microsoft.com

Enter destination domains or IP-addresses here. To separate them use space.
Example: mail.ru e-mail.ru yahoo.com 192.168.1.1

3. For all users, allow access to:

- Google websites
- CDAC websites
- Bing websites
- Gmail websites

Proxy filter SquidGuard: Target categories / Edit / Target categories 

[General settings](#) [Common ACL](#) [Groups ACL](#) [Target categories](#) [Times](#) [Rewrites](#) [Blacklist](#) [Log](#)

[XMLRPC Sync](#)

General Options

Name	common_whitelist Enter a unique name of this rule here. The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter.
Order	social_sites  Select the new position for this target category. Target categories are listed in this order on ACLs and are matched from the top down in sequence.
Domain List	google.com www.google.com gmail.com mail.google.com bing.com www.bing.com cdac.in www.cdac.in Enter destination domains or IP-addresses here. To separate them use space. Example: mail.ru e-mail.ru yahoo.com 192.168.1.1

Proxy filter SquidGuard: Target categories / Edit / Target categories ?

[General settings](#) [Common ACL](#) [Groups ACL](#) [Target categories](#) [Times](#) [Rewrites](#) [Blacklist](#) [Log](#)

XMLRPC Sync

General Options

Name

blacklist

Enter a unique name of this rule here.

The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter.

Order

blacklist



Select the new position for this target category. Target categories are listed in this order on ACLs and are matched from the top down in sequence.

Domain List

Enter destination domains or IP-addresses here. To separate them use space.

Example: mail.ru e-mail.ru yahoo.com 192.168.1.1

4. Allow access to social networking websites only for 3 hours (using a time-based access policy).

The screenshot shows the pfSense web interface for configuring a Proxy filter. The breadcrumb trail is "Proxy filter SquidGuard: Target categories / Edit / Target categories". The "Target categories" tab is selected in the top navigation bar. Below the navigation bar, there is a section for "General Options".

General Options

- Name:** social_sites. Below the input field, it says: "Enter a unique name of this rule here. The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter."
- Order:** common_whitelist. Below the dropdown, it says: "Select the new position for this target category. Target categories are listed in this order on ACLs and are matched from the top down in sequence."
- Domain List:** A text area containing the following domains: facebook.com, www.facebook.com, instagram.com, www.instagram.com, twitter.com, x.com, linkedin.com, www.linkedin.com, snapchat.com, www.snapchat.com. Below the text area, it says: "Enter destination domains or IP-addresses here. To separate them use space. Example: mail.ru e-mail.ru yahoo.com 192.168.1.1"

The screenshot shows the pfSense web interface for configuring a Proxy filter. The breadcrumb trail is "Proxy filter SquidGuard: Times / Edit / Times". The "Times" tab is selected in the top navigation bar. Below the navigation bar, there is a section for "General Options".

General Options

- Name:** social_time. Below the input field, it says: "Enter a unique name of this rule here. The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter."
- Values:** A table with two rows. Each row has a "Time type" dropdown (set to "Weekly"), a "Days" dropdown (set to "all"), a "Date or Date range" input field (empty), and a "Time range" input field. The first row's time range is "14:00-17:00" and the second row's is "00:00-23:59". Each row has a "Delete" button to its right.
- Add:** A green button with a plus sign and the text "Add".
- Description:** socia media only for 3 hour. Below the input field, it says: "You may enter any description here for your reference. Note: Example for Date or Date Range: 2007.12.31 or 2007.11.31-2007.12.31 or *.12.31 or 2007.*.31 Example for Time Range: 08:00-18:00".

At the bottom of the form, there is a blue "Save" button.

Package / Proxy filter SquidGuard: Times / Times

General settings Common ACL Groups ACL Target categories **Times** Rewrites Blacklist Log XMLRPC Sync

Name	Description	
social_time	socia media only for 3 hour	 
		 Add

 Save

a) Configure Common ACL target rules in SquidGuard.

Package / Proxy filter SquidGuard: Common Access Control List (ACL) / Common ACL

General settings **Common ACL** Groups ACL Target categories Times Rewrites Blacklist Log XMLRPC Sync

General Options

Target Rules

Target Rules List  

ACCESS: 'whitelist' - always pass; 'deny' - block; 'allow' - pass, if not blocked.

Target Categories		
[ditiss_whitelist]	access	allow 
[social_sites]	access	deny 
[common_whitelist]	access	allow 
[blacklist]	access	deny 
[dbda_whitelist]	access	allow 
Default access [all]	access	deny 

b) Create user groups in SquidGuard Groups ACL.

Package / Proxy filter SquidGuard: Groups Access Control List (ACL) / Groups ACL

General settings Common ACL **Groups ACL** Target categories Times Rewrites Blacklist Log XMLRPC Sync

Disabled	Name	Time	Description	
	dbda	social_time		 
	ditiss	social_time		 
				 Add

 Save

c) Create ACL rule for user 'ditiss' and assign whitelist permissions.

Name	ditiss																																																
Enter a unique name of this rule here. The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter.																																																	
Order	dbda																																																
Select the new position for this ACL item. ACLs are evaluated on a first-match source basis. Note: Search for a suitable ACL by field 'source' will occur before the first match. If you want to define an exception for some sources (IP) from the IP range, put them on first of the list. Example: ACL with single (or short range) source ip 10.0.0.15 must be placed before ACL with more large ip range 10.0.0.0/24.																																																	
Client (source)	'ditiss'																																																
Enter client's IP address or domain or "username" here. To separate them use space. Example: IP: 192.168.0.1 - Subnet: 192.168.0.0/24 or 192.168.1.0/255.255.255.0 - IP-Range: 192.168.1.1-192.168.1.10 Domain: foo.bar matches foo.bar or *.foo.bar Username: 'user1' Ldap search (Ldap filter must be enabled in General Settings): ldapsusearch ldap://192.168.0.100/DC=domain,DC=com?sAMAccountName=sub?(&(sAMAccountName=%s)(memberOf=CN=it%2cCN=Users%2cDC=domain%2cDC=com)) <i>Attention: these line don't have break line, all on one line</i>																																																	
Time	social_time																																																
Select the time in which 'Target Rules' will operate or leave 'none' for rules without time restriction. If this option is set then in off-time the second ruleset will operate.																																																	
Target Rules	ditiss_whitelist social_sites common_whitelist !blacklist !all [all]																																																
Target Rules List ACCESS: 'whitelist' - always pass; 'deny' - block; 'allow' - pass, if not blocked. <table><thead><tr><th colspan="3">Target Categories</th><th colspan="3">Target Categories for off-time</th></tr><tr><th colspan="3"></th><th colspan="3">If 'Time' not defined, this is column will be ignored.</th></tr></thead><tbody><tr><td>[ditiss_whitelist]</td><td>access</td><td>allow</td><td>[ditiss_whitelist]</td><td>access</td><td>---</td></tr><tr><td>[social_sites]</td><td>access</td><td>allow</td><td>[social_sites]</td><td>access</td><td>allow</td></tr><tr><td>[common_whitelist]</td><td>access</td><td>allow</td><td>[common_whitelist]</td><td>access</td><td>---</td></tr><tr><td>[blacklist]</td><td>access</td><td>deny</td><td>[blacklist]</td><td>access</td><td>---</td></tr><tr><td>[dbda_whitelist]</td><td>access</td><td>---</td><td>[dbda_whitelist]</td><td>access</td><td>---</td></tr><tr><td>Default access [all]</td><td>access</td><td>deny</td><td>Default access [all]</td><td>access</td><td>allow</td></tr></tbody></table>		Target Categories			Target Categories for off-time						If 'Time' not defined, this is column will be ignored.			[ditiss_whitelist]	access	allow	[ditiss_whitelist]	access	---	[social_sites]	access	allow	[social_sites]	access	allow	[common_whitelist]	access	allow	[common_whitelist]	access	---	[blacklist]	access	deny	[blacklist]	access	---	[dbda_whitelist]	access	---	[dbda_whitelist]	access	---	Default access [all]	access	deny	Default access [all]	access	allow
Target Categories			Target Categories for off-time																																														
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Do not allow IP-Addresses in URL	☐ To make sure that people do not bypass the URL filter by simply using the IP-Addresses instead of the FQDN you can check this option. This option has no effect on the whitelist.																																																

d) Create ACL rule for user 'dbda' and define access permissions.

Name	dbda																																
Enter a unique name of this rule here. The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter.																																	
Order	ditiss																																
Select the new position for this ACL item. ACLs are evaluated on a first-match source basis. Note: Search for a suitable ACL by field 'source' will occur before the first match. If you want to define an exception for some sources (IP) from the IP range, put them on first of the list. Example: ACL with single (or short range) source ip 10.0.0.15 must be placed before ACL with more large ip range 10.0.0.0/24.																																	
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Do not allow IP-Addresses in URL	☒ To make sure that people do not bypass the URL filter by simply using the IP-Addresses instead of the FQDN you can check this option. This option has no effect on the whitelist.																																

e) Enable and start SquidGuard URL filtering service.

The screenshot shows the pfSense web interface for the 'Proxy filter SquidGuard: General settings' page. The breadcrumb trail is 'Package / Proxy filter SquidGuard: General settings / General settings'. The 'General Options' section has the 'Enable' checkbox checked, with an 'Apply' button below it. A status bar at the bottom of this section shows 'SquidGuard service state: STARTED'. The 'LDAP Options' section includes fields for 'Enable LDAP Filter' (unchecked), 'LDAP DN' (with a hint to configure the DN), 'LDAP DN Password' (with a hint about password format), 'LDAP Cache Time' (set to 0), 'Strip NT domain name' (unchecked), 'Strip Kerberos Realm' (unchecked), and 'LDAP Version' (set to Version 3).

f) Configure client browser to use proxy server (192.168.25.5:3128).

Configure Proxy Access to the Internet

☐ No proxy

☐ Auto-detect proxy settings for this network

☐ Use system proxy settings

☒ Manual proxy configuration

HTTP Proxy Port

☒ Also use this proxy for HTTPS

HTTPS Proxy Port

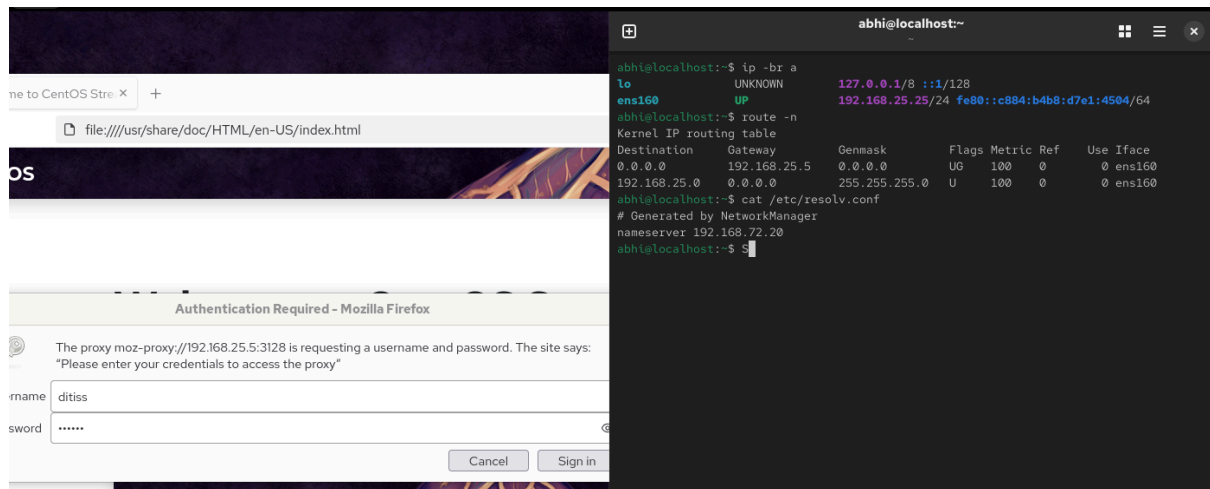
SOCKS Host Port

☐ SOCKS v4 ☒ SOCKS v5

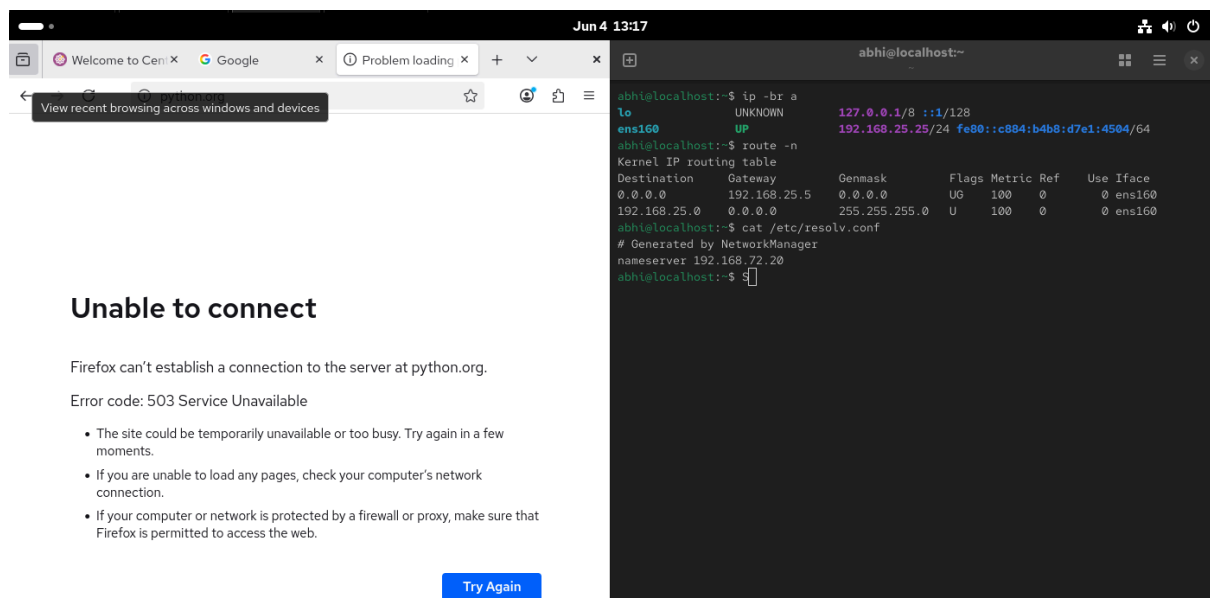
☐ Automatic proxy configuration URL

No proxy for

g) Authenticate to the proxy using user credentials.



h) Verify website access restrictions through SquidGuard filtering.



i) Configure Lightsquid reporting and generate proxy usage reports.

pfsense COMMUNITY EDITION System Interfaces Firewall Services VPN Status Diagnostics Help

Package / Squid Proxy Reports: Settings

Instructions

Perform these steps after install **IMPORTANT:** Click Info and follow the instructions below if this is initial install!

Web Service Settings

Lightsquid Web Port 7445
Port the lighttpd web server for Lightsquid will listen on. (Default: 7445)

Lightsquid Web SSL ☒ Use SSL for Lightsquid Web Access
This option configures the Lightsquid web server to use SSL and uses the WebGUI HTTPS certificate.

Lightsquid Web User admin
Username used to access lighttpd. (Default: admin)

Lightsquid Web Password
Password used to access lighttpd. (Default: pfsense)

Links [Open Lightsquid](#) [Open sqstat](#)

Report Template Settings

Language English
Select report language.

Report Template Base
Select report template.

Bar Color Orange
Select bar color.

j) View and analyze generated user web access reports in Lightsquid.

[Squid user access report](#)
Work Period: Jun 2026

Calendar
2026
01 02 03 04 05 06 07 08 09 10 11 12

Top Sites
YEAR YEAR YEAR
MONTH MONTH MONTH

Date	Group	Users	OverSize	Bytes	Average	Hit %
04 Jun 2026	grq	1	0	1.4 M	1.4 M	0.00%
Total/Average:		1	0	1.4 M	1.4 M	0.00%

LightSquid v1.8 (c) Sergey Erokhin AKA ESL

Not secure https://192.168.25.5:7445/user_detail.cgi?year=2026&month=06&day=04&user=192.168.25.25

[Squid user access report](#)
User: 192.168.25.25 (?)
Group: ?
Date: 04 Jun 2026

Total	Accessed site	Connect	Bytes	Cumulative	%
1	www.google.com:443	11	1.3 M	1.3 M	95.3%
2	192.168.25.5	14	67.574	1.4 M	4.6%
3	error:transaction-end-before-headers	7	0	1.4 M	0.0%
4	https:443	753	0	1.4 M	0.0%
Total			1.4 M		

LightSquid v1.8 (c) Sergey Erokhin AKA ESL